



OPEN Deep hierarchical networks for sentiment analysis of restaurant reviews from food apps

Md Humaion Kabir Mehedi¹, Fahmid Al Farid², Ehsanur Rahman Rhythm¹, Farhin Rahman¹, Khan Md Hasib^{3,4}, Jia Uddin⁵✉ & Sarina Mansor²✉

This study presents a framework for conducting sentiment analysis using deep learning techniques to evaluate restaurant food reviews collected from FoodPanda, a food delivery platform. We provide BDFoodSent, a unique large-scale dataset that includes over 334,000 reviews from FoodPanda in Bangladesh. These reviews were annotated for both binary (positive/negative) and multi-class (very bad, bad, neutral, good, and very good) sentiment classifications. Bag of Words (BoW) and TF-IDF feature extraction techniques were used to assess the traditional machine learning (ML) models. The models included decision trees, support vector machines, random forests, XGBoost, and k-nearest neighbors. To solve the issue of class imbalance, we used random oversampling, which resulted in a considerable improvement in the performance of the multi-class classification model. In addition, we present a hybrid deep learning architecture called H-Food. The design included embedding, BiLSTM, attention, and convolutional layers. The results of the experiments show that the proposed model achieved an accuracy of 91.42% and an F1-score of 91.35% when applied to binary classification tasks. Additionally, it achieved an accuracy of 78.74% and an F1-score of 78.72% when applied to multi-class classification using oversampled data. Comparative evaluations showed that the proposed architectures outperformed the classical models and baseline CNN design. The findings of this study demonstrate the efficacy of deep hierarchical networks in comprehending the feelings of users and provide insights that may be applied by restaurant services in emerging countries in the future.

Keywords Food reviews, Bangladeshi food, Restaurant review, Deep learning, NLP, Sentiment analysis

In the restaurant or food service business, a customer's opinion of a restaurant's food can make or break the business. Although Yelp or any food review platform was solely created to help food fanatics or people, in general, find a good place to consume good food, sometimes, reviews can be biased and misrepresent the restaurant's service. Sentiment analysis can be used to find and comb through a myriad of customer reviews on sites like Yelp, Trip Advisor, Uber Eats, Zomato, and Foodpanda, and many others to create an understanding of what a customer wants and what a restaurant might improve upon to get better reviews from said customers. This analytical approach is crucial, as evidenced by a comprehensive study on online food delivery services. A study¹ revealed that factors such as delivery efficiency, food quality, hygiene, and pricing greatly influence customer's satisfaction. These insights, derived from analyzing consumer behavior and preferences, underscore the importance of continuously monitoring and adapting to customer feedback. Restaurants and food delivery services can leverage such data to refine their offerings, address customer concerns more effectively, and enhance their reputation and success in the highly competitive food service industry.

Although reviews are important for determining the quality of a restaurant, they are not always the case, especially in developing countries. Bangladesh. Many factors, both external and internal, are attributed to a restaurant once a customer consumes their food. The foremost factors are bias, prejudice, and expectations, which can differ from one individual to another. Since COVID-19, the food delivery app "FoodPanda" has seen an increase as it was one of the first to provide contactless delivery service and it is still dominant in the

¹Department of Computer Science & Engineering, BRAC University, Dhaka, Bangladesh. ²Centre for Image and Vision Computing (CIVC), COE for Artificial Intelligence, Faculty of Artificial Intelligence and Engineering (FAIE), Multimedia University, Cyberjaya 63100, Selangor, Malaysia. ³Department of Computer Science & Engineering, Bangladesh University of Business and Technology, Dhaka, Bangladesh. ⁴School of Systems & Computing, The University of New South Wales, Canberra, ACT 2600, Australia. ⁵Artificial Intelligence and Big Data Department, Endicott College, Woosong University, Daejeon, South Korea. ✉email: jia.uddin@wsu.ac.kr; sarina.mansor@mmu.edu.my

fast food scene in Bangladesh due to its guerrilla marketing strategy and aggressive advertising in social media platforms like YouTube, Facebook, and others. According to The Business Standard, FoodPanda controls 65% of the fast-food market in Bangladesh, with 35,000 affiliated restaurants and over 80,000 riders. Hence, we can surmise that reviews on Foodpanda can have a huge impact on a restaurant business, much like Yelp does in first-world countries, due to the demographic shift towards urban consumers with busy lifestyles, leading to a higher demand for services such as Foodpanda². Furthermore, the radical increase in Internet usage has resulted in more food deliveries ordered by people of different age groups. Thus, we decided to use reviews from Foodpanda as our primary data in our research so that we may utilize a popular technique known as “sentiment analysis” with the help of different machine learning (ML) models and deep learning (DL) models to determine the sentiment behind reviews left by customers.

The key contributions of this study are as follows:

1. We created BDFoodSent, an exclusive large-scale labeled dataset of Bangladeshi restaurant reviews over 334k taken from Foodpanda, facilitating future studies in regional sentiment analysis.
2. We examine the impact of TF-IDF and BoW feature extraction methods on six prominent machine learning algorithms for sentiment classification.
3. We present H-Food, a hybrid deep learning architecture that integrates embedding, BiLSTM, attention, and CNN layers, designed for hierarchical sentiment analysis.
4. We assess our models on binary and multi-class tasks, with H-Food attaining 91.42% accuracy in binary classification and 78.74% accuracy in multi-class classification, surpassing baseline CNNs and conventional machine learning models.
5. We demonstrate that random oversampling successfully mitigates the class imbalance issue and enhances deep learning performance in fine-grained sentiment analysis tasks.

Literature review

A brief prelude is presented on the previous work of sentiment analysis involving reviews of restaurants. One of the studies³ concludes that SentiLSTM, a deep learning model, effectively analyzes sentiment in restaurant reviews. It is more precise in classifying reviews as positive, negative, or neutral compared to traditional models. By highlighting the critical aspects of reviews, the model helps businesses understand customer sentiments more precisely. It extracts essential features from reviews, such as service, food quality, and ambience, thereby enhancing the overall customer experience. This study proves that in complex review environments, deep learning models such as SentiLSTM outperform traditional methods. This is a step forward in leveraging AI for a more nuanced understanding of customer feedback. They used the Yelp dataset, which contained approximately 2000 restaurant reviews. In another study⁴, the same datasets were used; however, the authors combined an arcing classifier with other methods to create a hybrid model, and their accuracy was measured to determine how well they performed. Using naïve bayes (NB), support vector machines (SVM), and genetic algorithm (GA), a classifier ensemble was created and organized into the following groups: food, service, ambience, deals and discounts, and worthiness. Similar work was done⁵ using the available machine learning models such as random forest, XGBoost, logistic regression, and multinomial naïve Bayes to categorize the sentiments of book reviews instead of restaurant reviews, but the process and model build-up were similar. Moreover, this study presented a dataset with 204,659 book reviews in Bangla from two online bookshops, Rokomari and Wafilife. Additionally, they incorporated transformer-based and neural-based models for sentiment classification techniques into their analysis, which encompass long short-term memory networks (LSTM), global vectors for word representation (GloVe), and two iterations of the state-of-the-art Bangla-BERT model, namely, Bangla-BERT (base-uncased) and the more extensive Bangla-BERT (large), which usually performs better for large text corpora and semantic similarity and textual entailment, and also in multilingual cases, as the comments were both in Bangla and English with an accuracy of 96.7%. One of the most prominent datasets used by restaurants is customer reviews of Zomato, an Indian multinational food delivery company. One study⁶ used 150,000 reviews from Zomato Bangalore, which were categorized as positive (rating above 3), negative (rating below 3), and neutral (rating of exactly 3). As they worked with word corpora, some data preprocessing was common, including cleaning and converting the text into a structured format suitable for analysis. The steps involved converting all text to lowercase, tokenization (breaking text into individual words), removing punctuation and stop words, and using the TF-IDF Vectorizer to convert text into numerical data. Another study⁷ examined how advanced computer models, including machine learning, deep learning, and techniques that make these models more understandable, are used to determine customer perceptions of food delivery services. It discusses the rise in these services during the pandemic and the value of customer feedback in evaluating companies' performance. This paper reviews the existing research in this area, noting the common use of certain methods and the need for more transparent and interpretable models. This suggests that future research should focus on using deep learning to understand customer opinions on food delivery, along with methods that make these models clearer, such as LIME and SHAP. This study highlights the importance of making these complex models accurate and comprehensible. Recently, there has been a surge in online user content, posing challenges in personalizing services and extracting useful information. To address this, a study was conducted⁸ that utilized a bidirectional gated recurrent unit (Bi-GRU) approach in sentiment analysis, which is advantageous because of its efficient weight updates and context consideration and requires fewer parameters. This study developed a sentiment analysis model that integrates Word2vec, Bi-GRU, and attention deep learning techniques trained on Dianping.com restaurant reviews. The model achieved an F1-score of over 89%, outperforming the standard models. However, the imbalance in review sentiments and neglect of review polarity are limitations, suggesting future improvements with clustering and advanced machine learning techniques.

We now consider a few studies that used DL based models for consumer sentiment analysis using a hybrid feature extraction approach. They¹² used datasets such as SemEval-2014, Sentiment140, and STS-Gold, and then combined review-related features (RRF) and aspect-related features (ARF) to form a distinctive feature vector for each review. For RRF, they focused on term frequency-inverse document frequency (TF-IDF), n-gram features, and emoticon polarities to represent the emotional aspects of reviews. However, for ARF, the extracted aspect terms and their polarities were extracted using co-occurrence frequencies, addressing issues such as sarcasm and ambiguity in reviews. Finally, sentiment classification was performed using a long short-term memory (LSTM) deep learning classifier. This part of the model addresses the vanishing gradient problem and enhances learning from long data sequences. The model was experimentally evaluated using the mentioned datasets, compared with state-of-the-art methods, and achieved an accuracy of approximately 95.9 %. Similarly, another¹⁰ deep learning approach used a multiple-attention mechanism within a recurrent neural network (RNN), enhancing the model's ability to focus on relevant sentiment features in complex sentences. The novelty of this study is that it addresses the limitations of traditional sentiment analysis methods, which struggle with distant or subtle sentiment expression. They worked with multiple datasets, including restaurant reviews from SemEval2014. The use of weighted memory and multiple attention mechanisms enhances the accuracy and granularity of sentiment analysis. Another study¹³ includes a comparative analysis of BERT, with an accuracy of 92.86% against other techniques such as char-CNN, graph-CNN, LSTM, and Bi-LSTM. They did not technically use Foodpanda reviews as we did for our research; rather, they used reviews on Facebook, where most of the datasets were collected from the Foodpanda Facebook page.

In Table 1 a summary of the previous dataset information is provided.

Deep learning has been widely used in most natural language processing tasks, particularly in tasks such as text mining and sentiment analysis. Previous studies, such as analyzed consumer reviews, extracted opinions, and predicted product ratings, were conducted by employing recurrent neural networks (RNNs), convolutional neural networks (CNNs), and transformer-based models. For instance, researchers have proven the effectiveness of BERT and its multiple variants in capturing contextual meaning, which leads to significant improvements in classification accuracy compared to traditional machine learning techniques. Likewise, when entering the specialized context of food and healthcare reviews. It has been seen that domain-specific embedding and hybrid models that combine lexicon-based and deep learning methods have been shown to improve sentiment analysis performance.

In addition to text-based applications, deep learning has also achieved remarkable success in image and biomedical signal analysis, demonstrating its versatility across domains. For instance, it has been demonstrated that a lightweight CNN optimized by genetic algorithms can achieve an accuracy of over 99% in the classification of almond varieties. As a result, it is appropriate for evaluating agricultural quality in real time¹⁴. Additionally, the ROPGCViT model is an explainable transformer-based architecture for medical imaging in ophthalmology that encompasses attention and residual modules, achieves 94.7% accuracy for retinopathy of prematurity, and provides clinical interpretability through Grad-CAM visualizations¹⁵. Similarly, the MaxGlaViT framework, based on multi-axis vision transformers with enhanced attention mechanisms, achieved 92% accuracy in diagnosing glaucoma stages¹⁶. Deep learning has also been applied to hematology outside ophthalmology. The BC-YOLO model, which added MBConv-ECA blocks to a YOLOv11 backbone, allowed for the precise real-time identification of red blood cells, white blood cells, and platelets from microscopic images¹⁷. Collectively, these new studies demonstrate the resilience of CNN and transformer architectures to different kinds of data, which strengthens the case for and necessity of using deep learning in our work.

Dataset
Data collection

The dataset used in this study was collected through web scraping from the FoodPanda Bangladesh site. <https://foodpanda.com.bd> shown in Fig. 1. After obtaining a list of restaurants from FoodPanda Bangladesh, the next step was to obtain detailed reviews of the restaurants. This study used anonymous user reviews. The review extraction process was conducted methodically using Python scripting and HTTP GET queries, allowing us to construct a large dataset encompassing many aspects of user experience and restaurant properties. The information gathered covers a wide range of topics, with 19 separate fields covering various aspects. These variables include necessary identifiers, such as 'uuid' for unique identification; temporal markers, such as 'createdAt' and 'updatedAt' to track the review's chronological data; and textual components that include the actual review text. Furthermore, the dataset includes categorical variables such as ratings_overall and labels, thus providing a comprehensive platform for our study. The combination of these areas not only assures a thorough grasp of customer feedback but also allows for more detailed evaluations of the elements driving restaurant performance and customer

Ref.	Dataset Name	No of Labels	Language	Number of Records
³	Yelp Dataset (2000 Restaurant Reviews)	2	Bengali	8,435
⁹	Yelp Dataset Challenge for experimental results	3	English	-
⁶	Reviews from Zomato	3	English	150,000
⁸	Dianping.com	2	Chinese	35,248
¹⁰	Laptop Reviews, Restaurant Reviews, Tweets, News Comments	3	English	66,833
¹¹	Facebook pages, review sections of different food delivery companies	3	Bengali	5,680

Table 1. Dataset Information of Previous Works.

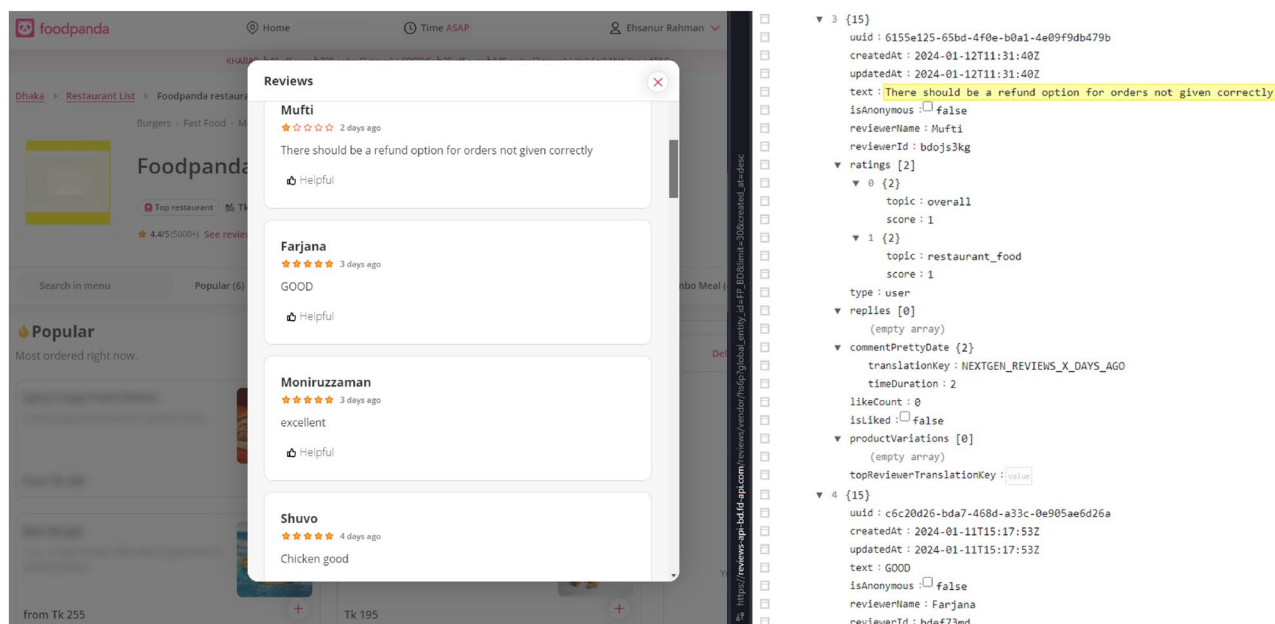


Fig. 1. Data scraping from FoodPanda by HTTP GET request.

id	createdAt	updatedAt	text	isAnonymous	reviewerName	reviewerId	ratings_overall	labels	ratings	type	likeCount	name	city	primary_cuisines	review_nu	restaurant	vertical_type	ids
68e8769	2023-11-02	2023-11-02	too much small amount	FALSE	User 1258	bdaclwhn	2	1	[{"topic": "overall", "score": 1}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
706a1e43	2023-10-2	2023-10-2	very small in amount	FALSE	User 1258	bdaclwhn	2	1	[{"topic": "overall", "score": 1}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
16cc99a3	2023-10-2	2023-10-2	food average but their desert box	FALSE	User 1945	bdjrmq	3	2	[{"topic": "overall", "score": 2}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
61a9f03	2023-10-2	2023-10-2	fresh and tasty	FALSE	User 6465	bdpr8of	5	3	[{"topic": "overall", "score": 3}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
a7835cdf	2023-10-1	2023-10-1	everything i ordered was good	FALSE	User 6057	bddrkjrd	4	3	[{"topic": "overall", "score": 3}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
7995071	2023-10-1	2023-10-1	kalo jira vhorta balu vhorta not r	FALSE	User 6057	bddrkjrd	4	1	[{"topic": "overall", "score": 1}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
547974ca	2023-10-1	2023-10-1	quantity of dal should be little m	FALSE	User 6057	bddrkjrd	4	3	[{"topic": "overall", "score": 3}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
4d7d419	2023-10-0	2023-10-0	beef was too poor in size quantiti	FALSE	User 4359	bd6r7ad	3	2	[{"topic": "overall", "score": 2}]	'user'	1	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
79e7563	2023-09-2	2023-09-2	chicken was not up to the mark r	FALSE	User 7338	bdrfk8ug	5	3	[{"topic": "overall", "score": 3}]	'user'	1	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
12854429	2023-09-2	2023-09-2	lot of sand in kali jeers bortha	FALSE	User 1190	bdwvw2zc	2	3	[{"topic": "overall", "score": 3}]	'user'	2	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
e79c98c	2023-09-1	2023-09-1	bad	FALSE	User 1190	e79c98c	1	1	[{"topic": "overall", "score": 1}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
ae740120	2023-09-0	2023-09-0	very good food i chose this home	FALSE	User 1896	ae740120	5	3	[{"topic": "overall", "score": 3}]	'user'	2	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
ab7255de	2023-09-0	2023-09-0	honestly eating today s beef ribs	FALSE	User 2499	ab7255de	5	3	[{"topic": "overall", "score": 3}]	'user'	1	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
0ce794cc	2023-09-0	2023-09-0	ate good food after a long time	FALSE	User 2499	0ce794cc	5	3	[{"topic": "overall", "score": 3}]	'user'	1	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
f51ab9de	2023-08-2	2023-08-2	really good clean tasting will ord	FALSE	User 11481	f51ab9de	5	3	[{"topic": "overall", "score": 3}]	'user'	1	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
da0d890e	2023-08-2	2023-08-2	too spicy with dry chilli	FALSE	User 7425	da0d890e	1	1	[{"topic": "overall", "score": 1}]	'user'	1	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
ca72630	2023-08-2	2023-08-2	good quality	FALSE	User 3636	ca72630	5	3	[{"topic": "overall", "score": 3}]	'user'	1	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
7be51abe	2023-08-1	2023-08-1	very good packaging now food w	FALSE	User 2659	7be51abe	5	3	[{"topic": "overall", "score": 3}]	'user'	1	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
5ac8eb30	2023-08-1	2023-08-1	fourth of its displayed	FALSE	User 2269	5ac8eb30	1	1	[{"topic": "overall", "score": 1}]	'user'	1	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
7d7f73c	2023-08-1	2023-08-1	alu bharta shukhi bharta dheral	FALSE	User 5101	7d7f73c	3	2	[{"topic": "overall", "score": 2}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
o6b674d2	2023-08-1	2023-08-1	bot was not so tender	FALSE	User 6577	bd0fssu	3	2	[{"topic": "overall", "score": 2}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
92ccca70	2023-08-0	2023-08-0	very bad delivery person	FALSE	User 1389	92ccca70	1	1	[{"topic": "overall", "score": 1}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
2e33b555	2023-08-0	2023-08-0	amount of food is poor compari	FALSE	User 5592	bd82c4z	2	1	[{"topic": "overall", "score": 1}]	'user'	1	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
8dcdeff-b	2023-08-0	2023-08-0	teast is good thanks	FALSE	User 2441	8dcdeff-b	3	3	[{"topic": "overall", "score": 3}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
6cd07169	2023-07-0	2023-07-0	spicy egg fry chilo dey nail r dher	FALSE	User 14841	n4gs44p	1	1	[{"topic": "overall", "score": 1}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
5c9524fc	2023-07-0	2023-07-0	best food always	FALSE	User 13651	5c9524fc	5	3	[{"topic": "overall", "score": 3}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
9d0557c1	2023-07-0	2023-07-0	good food	FALSE	User 2659	9d0557c1	5	3	[{"topic": "overall", "score": 3}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
9377dbb3	2023-07-0	2023-07-0	very good standard of foods the	FALSE	User 5603	9377dbb3	5	3	[{"topic": "overall", "score": 3}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
6b07d95f	2023-07-0	2023-07-0	quantity of food is very bad	FALSE	User 7983	bdrh9ss	3	2	[{"topic": "overall", "score": 2}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
a120419e	2023-06-2	2023-06-2	the quantity and the pieces of be	FALSE	User 2659	k6uz8gq	1	1	[{"topic": "overall", "score": 1}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_based_kitchen']	
f060f060	2023-06-0	2023-06-0	best homemade food in uttara a	FALSE	User 1367	f060f060	5	3	[{"topic": "overall", "score": 3}]	'user'	0	Restaurar Dhaka	Curry	['Curry', 'R']	146	3.7	['restaurants', 'home_basee_kitchen']	

Fig. 2. Collected Dataset in CSV format.

satisfaction in the future. This comprehensive and complex dataset is an excellent resource for performing in-depth research and gaining significant insights into restaurant ratings and user preferences. We have hosted our dataset in Zenodo <https://zenodo.org/records/14259962>.

Dataset overview

Following preprocessing, our dataset had a remarkable size, consisting of 334,119 rows and 19 columns. This large dataset offers a rich environment for in-depth investigation and has the potential to yield a thorough understanding of the dynamics of restaurant ratings in Bangladesh.

In addition to the size of the dataset, the textual portion was statistically examined. The user-generated reviews exhibited a range of sentence lengths, with a minimum sentence length of 3 and a maximum sentence length of 366. These statistical discoveries play a critical role in determining how we will tackle future deep learning and natural language processing assignments.

We implemented strong quality assurance methods. This was achieved by cross-checking the translated text with the original text to ensure the fidelity of the translation. To ensure accuracy and coherence, the sentiment labels were subjected to the same examination. Figure 2 shows the data collection in CSV format.

In summary, our dataset, obtained from the FoodPanda Bangladesh website, underwent rigorous extraction, preprocessing, and quality assurance. It serves as a solid framework for the following aspects of our research, in which we investigated the use of deep learning techniques to unravel the complicated tapestry of sentiments hidden in restaurant reviews in Bangladesh.

Ethical considerations and uniqueness

Anonymization: No information that could be used to identify a user, like usernames, emails, or order IDs, was collected or stored.

Respect for Terms of Service: FoodPanda's publicly available data was used for scraping, and usage rates were limited to prevent server overload.

Use of Public Data: All of the reviews that were collected can be seen by anyone and are okay to use for research purposes.

Data Hosting: The final cleaned dataset is hosted on Zenodo under the CC BY 4.0 license so that it can be used again.

In contrast to existing datasets such as Yelp, which is primarily US-based, English-only, and often structured, and Zomato, which is region-limited and lacks code-mixed data, BDFoodSent offers a distinctive low-resource, underrepresented regional corpus from Bangladesh. It features rich code-mixed (Banglish) language content and a higher volume of reviews (334 K +) than many previous domain-specific datasets. Additionally, the cultural context of the reviews reflects localized expressions, expectations, and sentiments unique to Bangladeshi food services. Consequently, BDFoodSent serves as an indispensable resource for multilingual natural language processing (NLP), code-mixing research, and region-specific sentiment analysis in developing contexts.

Methodology

In this research, we followed the steps presented in Fig. 3.

Data preprocessing

Understanding the importance of high-quality data, we carefully preprocessed the dataset. To strengthen the integrity of the dataset and provide a solid base for our study, null rows were removed. Additionally, the textual data were translated from Bangla and Banglish into English to provide a consistent language for further analysis.

A crucial component of our preprocessing was the ratings_overall column sentiment labeling. Together, ratings 1 and 2 were designated as negative (1), rating 3 as neutral (2), and ratings 4 and 5 as positive (3). In Fig. 4 shows data distribution between positive, negative and neutral class. We considered positive and negative classes for binary sentiment classification, as neutral was the minority class. In contrast, Fig. 5 shows data distribution for 5 (very bad=1, bad=2, neutral=3, good=4, and very good=5) types of sentiment. The multi-class categorization expedited the sentiment analysis stage and offered a more straightforward structure for future research. Even after that, extensive preprocessing was required to clean the data for further analysis. The following steps were performed:

1. **Text Normalization:** The reviews were converted to lowercase using the Python lower() method to ensure uniformity. Abbreviations and spelling variations were mapped to their standard forms using regular expressions. For example, “don’t” became “do not,” “kebab” became “kabab,” etc.
2. **Noise Removal:** To reduce clutter, non-alphabetic characters such as digits, punctuation, tags, and excess whitespace were eliminated utilizing regular expressions.
3. **Translation:** Employing automated machine translation tools, the normalized Bengali and Banglish reviews were converted to English. This was a crucial stage because the sentiment analysis models required the reviews to be written in English.
4. **Quality Checks:** Spot checks were conducted manually through the process of cross-verifying translated evaluations with the original texts. This aided in validating the accuracy and integrity of the translation.
5. **Sentiment Labeling:** The translated reviews were labeled with sentiment tags (positive, negative, neutral) based on the ratings_overall field, simplifying the classification task.

Raw datasets often require extensive preprocessing, particularly when they are sourced from non-English contexts. The systematic cleaning, translation, and quality checks implemented helped construct a high-quality translated dataset, preserving the integrity of the original reviews while transforming them into a standard format for further analysis. To maintain the dataset's completeness, supplementary checks were implemented to detect and correct any unintentional omissions, guaranteeing that the dataset contained all data obtained from FoodPanda Bangladesh.

Feature extraction

TF-IDF

For every word in a document, TF-IDF assigns a number based on the inverse relationship between the word frequency in that document and the percentage of documents in which it occurs¹⁸. TF-IDF scores are closely tied to specific documents, indicating that if such a word is used in a search, the document may be relevant to the user. This method helps identify keywords and gauge their significance in the documents. A statistical technique mostly used in NLP, it shows the relative relevance of words based on their frequency of occurrence in the text corpus. The “Term Frequency” is the number of times a word occurs in the document if more frequent, higher TF value. The Inverse Document Frequency (IDF) calculates the rarity of a word in a document, rarely exceeding the IDF value. These can be used to understand the relevance of a word, alongside understanding its significance in the corpus.

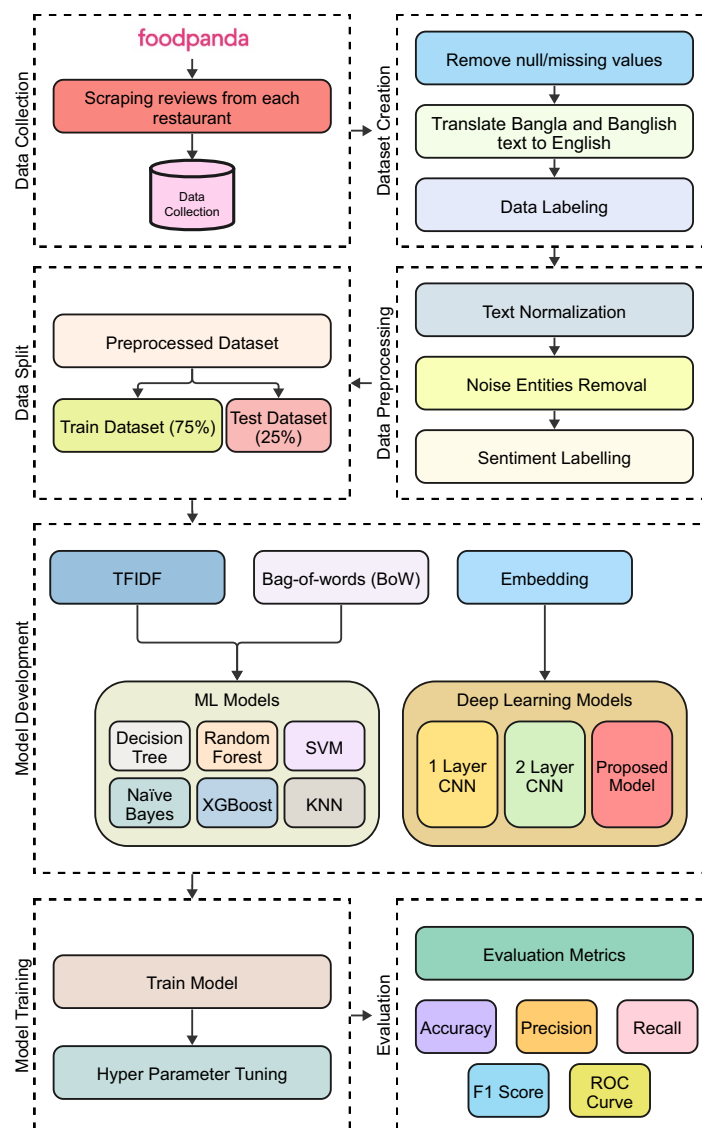


Fig. 3. Workflow of the proposed architecture.

Bag of words (BoW)

The bag of words model¹⁹ is a fundamental technique for text representation that discards the sequential order of words and focuses solely on their occurrence in a document. First, the text was broken down into individual words or tokens. Subsequently, the vocabulary of unique words in the dataset was compiled. Vectorization is performed, where each element corresponds to the count of a specific word in the vocabulary, representing each document as a numerical vector.

Vocabulary Creation: Compiling a vocabulary of unique words present in the entire dataset.

Vectorization: Each document is represented as a numerical vector, where each element corresponds to the count of a specific word in the vocabulary.

The resulting vectors were high-dimensional and sparse, capturing word frequencies across documents. Despite its simplicity, the BoW model is a foundational feature extraction technique for various text-based machine learning tasks.

Word embedding

Word embedding bridges the gap between computers and language²⁰. This NLP method identifies the relationships and meanings of words in a high-dimensional space by assigning numerical vectors to the words. Words that indicate proximity are indicative of semantic similarity. This gives computers the capability to assess text by means other than mere word displays, allowing for many applications, such as categorization, translation, and even sophisticated conversation. Word embedding transforms our understanding and interaction with language by converting meanings into numerical representations. We converted the text into a numeric number

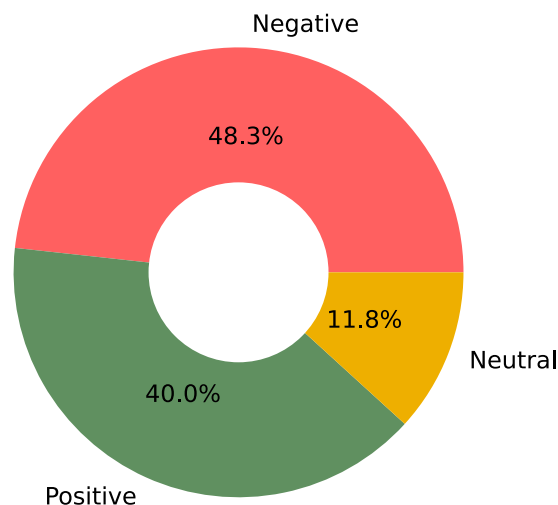


Fig. 4. Data distribution between positive, negative and neutral class.

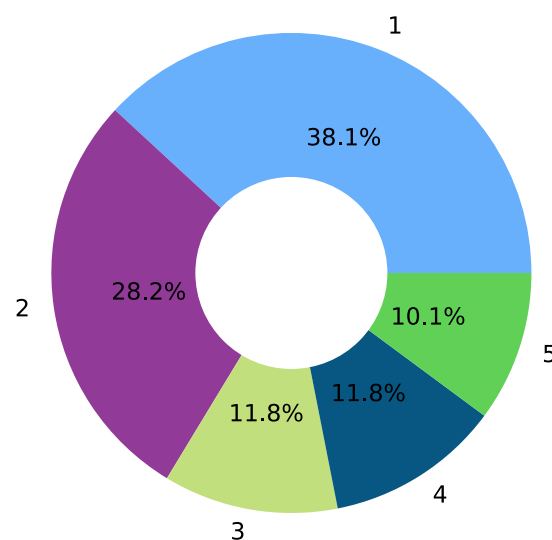


Fig. 5. Data distribution of 5 types of sentiment (very bad, bad, neutral, good, and very good).

using the `text_to_sequence()` method. Then, post-padding was added with `maxlen=72`, as after preprocessing, the max length of the sentence was 72 and the min length was 3.

Random oversampling

Random oversampling was used in this study to address class imbalance in the sentiment analysis dataset. Random oversampling is a technique for balancing the class distribution of an imbalanced dataset by duplicating the examples from the minority class. This method aims to increase the diversity and representativeness of the minority class while preserving information in the original data. Because certain sentiment categories, such as “very bad,” had a small number of samples compared to more frequent classes, such as “good,” directly training machine learning models can lead to bias.

In this study, we applied random oversampling to the training dataset, which consisted of 334,119 restaurant reviews with different sentiment ratings, as shown in Fig. 6. We randomly selected and duplicated examples from the minority classes until they reached the same number as the majority class. Subsequently, we used this dataset to train different machine and deep learning models for sentiment analysis. We then checked the performance of these models by comparing their results with those obtained using the original dataset. Through

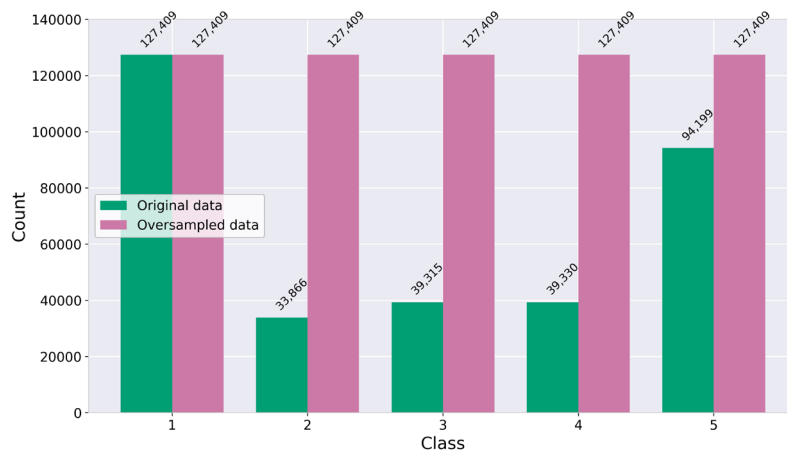


Fig. 6. Original vs oversampled data.

this balanced class distribution, the models could be trained without any bias towards more frequent categories. The results indicate major progress in performance metrics, such as accuracy, precision, recall, and F1-score, owing to oversampling, particularly for the multi-class sentiment classification. By alleviating class imbalance, the models could effectively learn the predictive patterns for all sentiment ratings.

Data split

In this study, the dataset underwent various data-splitting strategies to evaluate model performance. Three different splits were employed. In the first 90:10 configuration, 90% of the data were allocated for training the models, while the remaining 10% were reserved for testing. This split provides a larger training set, potentially aiding the models in learning complex patterns. Second, the 80:20 dataset was divided into 80% for training and 20% for testing. This split strikes a balance between a substantial training set and a sizable testing set, allowing for a more comprehensive assessment of model generalization. Finally, a 75:25 split, where 75% of the data was designated for training and 25% for testing, emerged as the optimal configuration. This split yielded the best outcomes because it strikes a balance between a sufficiently large and robust training and testing set. These varied data-splitting approaches enabled a thorough evaluation of the model performance under different conditions, shedding light on their ability to generalize and make accurate predictions across diverse datasets. The 75:25 split was particularly effective, emphasizing the importance of balancing the proportions of the training and testing data.

Model implementation

Decision Tree (DT)

Decision trees are versatile tools in machine learning that efficiently perform both classification and regression tasks. Unlike some methods that require the consideration of multiple features, decision trees categorize data using a single decision step. They organize decisions hierarchically, forming a tree structure with a root node (data), internal nodes (splits), and terminal nodes (exterior nodes)²¹. Each node makes a binary choice, segregating classes, with processing concentrated in top-down traversal, ending in leaf nodes. In decision tree regression, the model approximates real-valued functions. Based on binary recursive trees, the construction iteratively partitions data into segments, adeptly handling large datasets without complex parametric frameworks. Training and validation datasets were created after the sample threshold was reached. Training builds the model, and validation evaluates its performance and determines the optimal tree size²².

Random Forest (RF)

Decision trees are combined to form random forests, which are widely used for load demand forecasting. Regression trees precisely predict load demand by iteratively breaking down training data²³. Regressive decision trees aim to achieve accurate load predictions by segmenting the data further if the prediction errors exceed the thresholds. The evaluation employs the Mean Squared Error, calculating the overall tree error through a weighted average of node errors. The general equation for the Random Forest prediction aggregates the individual decision tree outputs. Each tree contributes to the final prediction, averaging for regression tasks or majority vote for classification. This ensemble methodology mitigates overfitting and enhances the model accuracy and stability by constructing multiple decision trees during training, each using a random subset of the dataset.

Support Vector Machine (SVM)

Support Vector Machines (SVMs) are gaining popularity in machine learning and pattern classification. They create a separation boundary in the input space, linear or nonlinear, by combining kernels associated with Support Vectors²⁴. SVMs can be utilized for classification and regression tasks, creating an optimal hyperplane based on the proximity of features and distribution of data points. For nonlinear relationships, the kernels map the data into higher-dimensional spaces. SVMs establish a margin around the hyperplane, influencing the

sensitivity to misclassifications via the C parameter²⁵. They excel in high-dimensional environments and can handle linear and nonlinear interactions. Careful selection of kernels and parameter adjustments are crucial for optimal results. The decision function for the SVM is expressed as a weighted sum of the similarities between the input and support vectors, plus a bias term. The kernel choice shapes the decision boundary, with common types including linear, polynomial, and radial basis function (RBF) kernels. The SVM aims to identify an effective hyperplane that accurately distinguishes the data points in high-dimensional feature spaces.

Naive Bayes (NB)

The Bayes theorem remains fundamental to sophisticated machine learning models and inferential statistics. It updates the probabilities of hypotheses based on new information, utilizing Bayesian reasoning to account for prior knowledge²⁶. Bayesian analysis enables addressing topics beyond the scope of frequentist methodologies, which lack the concept of assigning probabilities to hypotheses²⁷. Naive Bayes comprises three types: Gaussian Naive Bayes for real-valued features, Multinomial Naive Bayes for discrete features like word counts, and Bernoulli Naive Bayes for binary features. Naive Bayes is a probabilistic classification algorithm whose formula is based on Bayes' theorem. The posterior probability of a class-given feature is proportional to the prior probability of the class multiplied by the likelihood of each feature, given the class. This reflects the core assumption of feature independence, given the class. The decision process entails selecting the class with the highest probability after considering all available information to be the predicted class.

XGboost (XGB)

XGBoost, or extreme gradient boosting, has gained prominence in machine learning because of its efficacy. Using an ensemble learning algorithm, XGBoost effectively amalgamates predictions from numerous weak learners²⁸, typically, decision trees. Its methodology involves systematically constructing a series of decision trees, where each subsequent tree rectifies the errors made by its predecessors, collectively refining the overall model design. XGBoost is distinguished by its adept utilization of gradient boosting and learning from previous errors to progressively enhance predictive accuracy. This continuous fine-tuning process makes XGBoost a reliable and adaptable tool suitable for refining predictions across various machine learning scenarios. Its sequential and corrective nature renders it particularly valuable for improving the accuracy and reliability of predictive modeling.

K-Nearest Neighbor (KNN)

The k-nearest neighbors algorithm is valuable for classification and regression tasks²⁹. It classifies a data point by considering the majority class among its k nearest neighbors based on the principle that similar instances are likely to belong to the same class. KNN's simplicity of KNN makes it reliable, especially when neighboring instances significantly influence the predictions. It operates like a friendly neighbor, deriving cues from nearby cases to make predictions and acknowledging that proximity often implies a similarity. KNN's ease of use and intuitive philosophy of KNN make it a preferred choice in scenarios that prioritize simplicity and effectiveness.

Proposed model

Fig. 7 shows that our proposed model is a combination of embedding layer, BiLSTM layer, convolution layer and dense layers. Table 2 shows the summary parameter of the model.

- **Embedding layer:** Natural language processing (NLP) activities are the most common use cases for the embedding layer. It is widely recognized that textual data must be converted to numerical data when feeding them into neural networks or any other machine learning model. Each word can be converted into a vector of a predetermined size using an embedding layer. The embedding layer has three primary parameters.

1. **input_dim** : Size of the vocabulary
2. **output_dim** : Length of the vector for each word
3. **input_length** : Maximum length of a sequence

- **BiLSTM layer:** Bidirectional LSTM (BiLSTM) comprises two unidirectional LSTMs that process the sequence in both the forward and backward directions simultaneously. Two independent long short-term memory (LSTM) networks may be shown in this³⁰ model; one receives the token sequence in its original order, while the other accepts it in reverse. The final output is the sum of the two probability vectors returned by these LSTM networks, each operating independently. It has three gates (input, forget, and output) and two states (cell and hidden states). The equations for the gates and states are as follows:

1. **Input gate:** $i_t = \sigma(W_{ii} \cdot x_t + b_{ii} + W_{hi} \cdot h_{(t-1)} + b_{hi})$
2. **Forget gate:** $f_t = \sigma(W_{if} \cdot x_t + b_{if} + W_{hf} \cdot h_{(t-1)} + b_{hf})$
3. **Output gate:** $o_t = \sigma(W_{io} \cdot x_t + b_{io} + W_{ho} \cdot h_{(t-1)} + b_{ho})$
4. **Cell state:** $c_t = f_t \cdot c_{(t-1)} + i_t \cdot \tanh(W_{ic} \cdot x_t + b_{ic} + W_{hc} \cdot h_{(t-1)} + b_{hc})$
5. **Hidden state:** $h_t = o_t \cdot \tanh(c_t)$

- **Convolutional layer:** Convolution is performed in a single dimension in a 1D convolutional layer³¹, which is usually used with sequences or time-series data. To implement the 1D convolutional layer, the following equations must be satisfied:

$$\text{– 1D Convolution Operation: } S(i) = (I * K)(i) = \sum_m I(m) \cdot K(i - m)$$

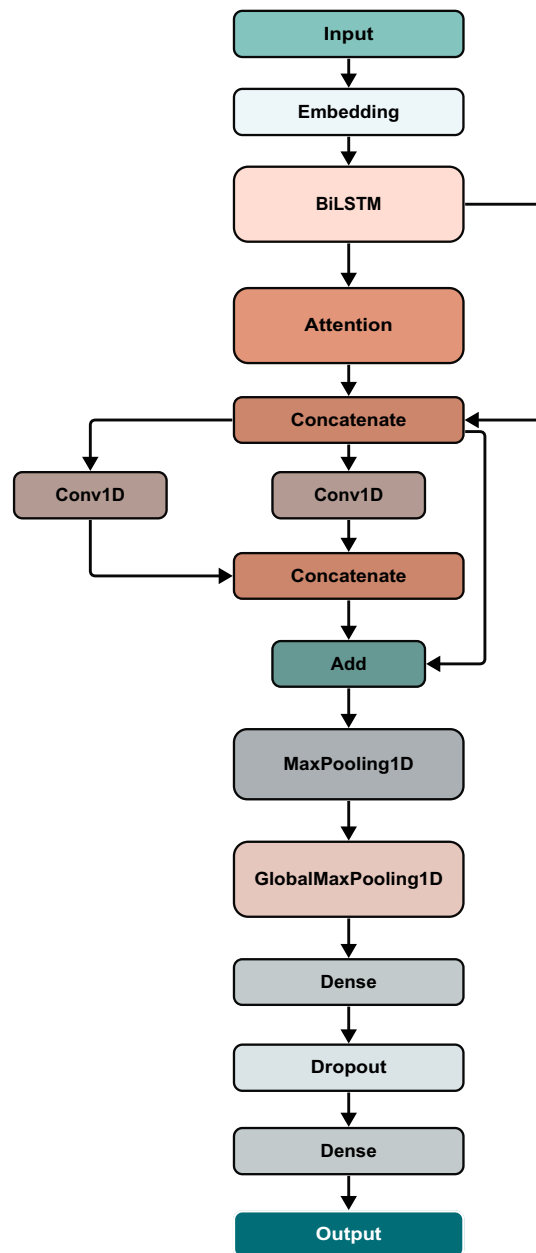


Fig. 7. Architecture of the Proposed Model.

Here, $S(i)$ is the value at position i in the feature map, I is the 1D input sequence, and K is a 1D convolutional filter.

- **1D Convolutional Layer output:** $C_o(i, k) = \sigma \left(\sum_m \sum_l I(m, l) \cdot W(i - m, k, l) + b_k \right)$

Here, $C_o(i, k)$ is the value at position i in the k -th feature map of the output, σ is the activation function (e.g., ReLU), $I(m, l)$ is the value at position m in the l -th channel of the input, W is a set of learnable weights, and b_k is the bias term for the k -th feature map.

- **Pooling layer:** Convolutional Neural Networks (CNNs) employ the pooling layer to downsample input feature maps, reduce computational complexity, and extract dominating features. Max pool and global max pooling are typical pooling methods.
 - **Max Pool:** Through the max pooling technique, the highest value from a predetermined area (pooling window) is extracted from the sequence.
Equation (for a region MP): $\text{MaxPooling}(MP) = \max_{i \in MP} (I(i))$
 - **Global Max Pool:** Through the global max pooling technique, the highest value from the entire area from the sequence.

Layer (type)	Output Shape	Param #
InputLayer	(None, 72)	0
Embedding	(None, 72, 300)	14400000
Bidirectional	(None, 72, 128)	186880
Attention	(None, 72, 128)	0
Concatenate	(None, 72, 256)	0
Conv1D	(None, 72, 128)	98432
Conv1D	(None, 72, 128)	163968
Concatenate	(None, 72, 256)	0
Add	(None, 72, 256)	0
MaxPooling1D	(None, 36, 256)	0
GlobalMaxPooling1D	(None, 256)	0
Dense	(None, 128)	32896
(Dropout	(None, 128)	0
Dense	(None, 64)	8256
Dense	(None, number of classes)	#
Total params: 14,890,432		
Trainable params: 14,890,432		
Non-trainable params: 0		

Table 2. Proposed Model Summary.

$$\text{Equation (from the entire sequence GAP): GlobalMaxPooling (GAP)} = \max_i(I(i))$$

- **Fully connected layer:** A dense layer, often called a fully connected layer, is a type of neural network architecture in which each neuron or node in the layer is linked to all the neurons in the layers below and above it³². The dense layer is used to classify the input image or text into labels. The role of this layer is to connect the convolution and pooling layers with the output layer while assigning labels to the inputs.

The H-Food model employs bidirectional long short-term memory (BiLSTM), attention mechanisms, and multi-scale convolutional layers to extract contextual and local features from restaurant review texts. BiLSTM analyzes input sequences in both forward and backward directions, enabling the model to comprehend sentence dependencies from preceding and subsequent words. In sentiment analysis, where word meanings frequently rely on context, a conventional LSTM, which processes sequences unidirectionally, fails to capture this bidirectional context adequately. Comprehending phrases such as “not very tasty” or “good but overpriced” requires an awareness of the context surrounding sentiment-laden adjectives. LSTM may overlook reverse dependencies that substantially influence emotion polarity. An attention mechanism was incorporated into the BiLSTM outputs to enhance interpretability and performance by emphasizing sentiment-laden phrases by weighting more informative tokens. This emphasis is effective in longer mixed-opinion evaluations when a few phrases capture sentiments. Multiple 1D convolutional layers with different kernel sizes allow the extraction of local patterns at various granularities, including brief words (“too salty”) and more extensive assertions (“highly recommended for families”). The concatenation of these convolutional layer outputs generates a comprehensive feature representation derived from diverse n-gram data. The H-Food model employs BiLSTM for global sequential context, attention mechanisms for word-level significance, and multi-scale CNNs for localized pattern extraction, rendering it robust in detecting nuanced sentiment signals among different unstructured and code-mixed restaurant review data.

Evaluation metric

To comprehensively evaluate the performance of machine learning and deep learning models in sentiment analysis tasks, it is vital to assess them using appropriate metrics that quantify the different facets of classification proficiency.

This study included six principal evaluation metrics³³: accuracy, precision, recall, F1-score, confusion matrix, and receiver operating characteristic (ROC) curve. Each statistic offers distinct insights into the models' proficiency in classifying sentiments from restaurant reviews.

Accuracy measures the overall performance of a classification model by evaluating the percentage of correct predictions among all the predictions. It provides a simple and intuitive method for performance assessment.

$$\text{Accuracy} = \frac{T_P + T_N}{T_P + T_N + F_P + F_N} \quad (1)$$

Here, T_P , T_N , F_P and F_N are the numbers of true positives, true negatives, false positives, and false negatives, respectively.

Precision evaluates the model accuracy by determining the ratio of correct positive predictions to the total positive predictions. It measures the precision of the model in predicting the positive class.

$$Precision = \frac{T_P}{T_P + F_P} \tag{2}$$

Recall is the percentage of actual positive cases that the model correctly predicts to be positive. It evaluates the sensitivity of the model in detecting a positive class.

$$Recall = \frac{T_P}{T_P + F_N} \tag{3}$$

The F1-score balances both precision and recall by calculating their harmonic means. It provides a single metric for model performance.

$$F1 = \frac{2 * Precision * Recall}{Precision + Recall} = \frac{2 * T_P}{2 * T_P + F_P + F_N} \tag{4}$$

The ability of a classification model to distinguish between classes was measured using the receiver operating characteristic (ROC) curve. This shows how correctly identifying positive instances affects the incorrect identification of negative ones. An ideal model maximizes true positives and minimizes false positives, as shown by the ROC curve. Models with higher AUCs performed better in terms of classification. The ROC curves assess a model's ability to distinguish between positive and negative sentiments from review texts.

Results analysis

An extensive evaluation was conducted on the performance of several ML and DL models for the sentiment analysis of restaurant food reviews. The models were assessed on binary classification of positive vs. negative sentiment, as well as multi-class classification across five sentiment ratings from very bad to very good.

Two feature extraction techniques were tested for the ML models: term frequency-inverse document frequency (TF-IDF) and bag of words (BoW). TF-IDF calculates the relevance of words by considering their frequency in a document compared to the entire corpus. BoW simply uses word count.

Six ML algorithms were implemented: decision tree (DT), random forest (RF), support vector machine (SVM), naïve bayes (NB), XGBoost, and k-nearest neighbors (KNN). For deep learning, 1 and 2 layer deep convolutional neural networks (DCNN) and a proposed model with word embeddings were trained and tested.

The evaluation metrics included accuracy, precision, recall, F1-score, ROC, and confusion matrix. Accuracy measures the overall effectiveness of the model. Precision evaluates the percentage of correct positive predictions. Recall measures the percentage of accurately anticipated positive instances among the total number of positive instances. The F1-score is a balanced metric that represents the harmonic mean of precision and recall.

Table 3 lists the hyperparameter configuration details of the proposed model. To train the model effectively, several hyperparameters were adjusted based on the data and the tasks. The padding length was set to 72, which means that the model can process input sequences up to this length. The number of epochs was varied from 100 to 250 to determine the optimal number of training cycles. For binary classification we trained our model 100 epochs, for multi-class 150 epochs and 250 epochs was executed for multi-class oversampled data. A batch size of 32 was selected, which was neither too small nor too large for the study. The learning rate was fixed at 0.0001, which allowed for stable and smooth training of the model. Stochastic Gradient Descent (SGD) was used as the optimizer³⁴ because it is simple and efficient. Different activation functions, such as ReLU, sigmoid for binary, and softmax for multi-class, were applied to different layers according to their functions. The loss functions were chosen as binary/categorical cross-entropy for binary and multi-class classification tasks, respectively. Therefore, the hyperparameters were systematically tuned to achieve optimal model training.

Machine learning models

To classify reviews as either positive or negative sentiment, TF-IDF enabled the ML models to achieve markedly higher accuracy and F1-scores than using BoW features in most cases. This indicates TF-IDF's greater efficacy of TF-IDF in extracting informative textual features for sentiment analysis.

Hyperparameters	Values
Padding length	72
Number of epochs	100,150,250
Batch size	32
Learning rate	0.0001
Optimizer	Sgd
Activation function	Relu, sigmoid, softmax
Loss function	Binary_crossentropy, categorical_crossentropy

Table 3. Hyperparameters configuration for Proposed model.

	ML Models	Accuracy	Precision	Recall	F1-score
Term Frequency-Inverse Document Frequency (TF-IDF)	Decision Tree	87.34	87.25	87.19	87.22
	Random Forest	80.59	84.07	78.94	79.33
	SVM	90.74	90.82	90.49	90.63
	Naive Bayes	87.43	87.59	87.03	87.23
	XGboost	89.72	90.13	89.15	89.50
	KNN	67.00	73.06	69.22	66.14
Bag of words (BoW)	Decision Tree	83.42	83.25	83.37	83.30
	Random Forest	81.18	83.55	79.78	80.21
	SVM	87.67	87.58	87.53	87.55
	Naive Bayes	86.60	86.44	86.64	86.52
	XGboost	86.03	86.21	85.58	85.80
	KNN	83.57	83.40	83.55	83.46

Table 4. Using TF-IDF and BoW binary (positive and negative) classification.

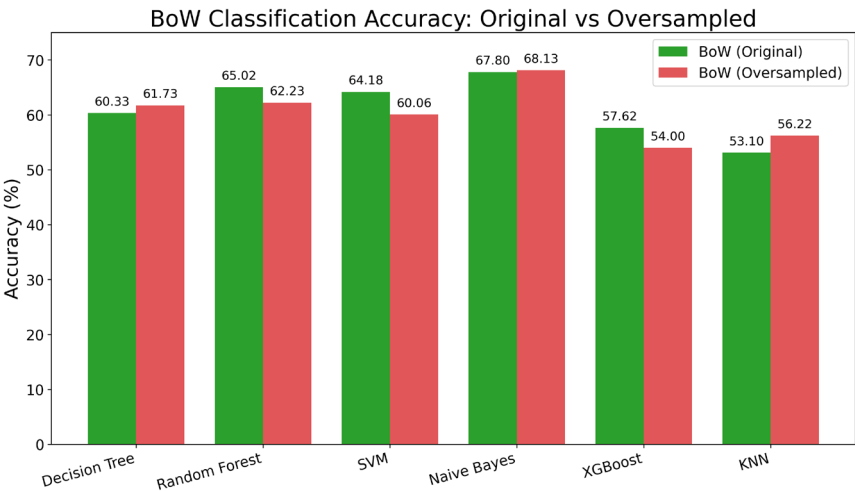


Fig. 8. Accuracy comparison of different machine learning models for multi-class sentiment classification using Bag of Words (BoW) features on the original dataset versus the oversampled dataset.

As presented in Table 4 among the ML models, SVM delivered the best performance with 90.74% accuracy and 90.63% F1-score using TF-IDF. In the case of BoW features, SVM achieved the highest accuracy of 87.67% and 87.55% F1-score. SVM creates an optimal separating hyperplane between classes, which appears to be effective for the binary separation of positive and negative sentiments. In contrast, KNN performed poorly compared to the other models, attaining only 67% accuracy with TF-IDF. This suggests that KNN is ineffective in identifying informative nearest neighbors for sentiment classification tasks. For multi-class prediction across five sentiment ratings, the performance of ML models was lower than that of binary classification, with an accuracy in the 44–65% range without oversampling. This decline highlights the difficulty in differentiating between finer-grained sentiment categories.

The set of four bar graphs depicted in Figs. 8, 9, 10, and 11 provides a detailed comparison of the performance of the machine learning model in sentiment classification tasks using Bag of Words (BOW) under two scenarios: without oversampling and with oversampled data.

In Fig. 8, Decision Tree achieves approximately 60.33% accuracy without oversampling and improves to 67.80% with oversampling. Logistic regression exhibited an accuracy of 65.61% without oversampling and 56.81% with oversampling. Random Forest shows an accuracy of 61.73% without oversampling and 68.13% with oversampling. SVM attained 65.02% accuracy without oversampling and 57.62% with oversampling.

Naive Bayes yielded an accuracy of 62.23% without oversampling and 54.00% with oversampling. XGBoost maintained an accuracy of 64.18% without oversampling and 53.10% with oversampling. KNN achieved an accuracy of 60.06% without oversampling and 56.22% with oversampling. The patterns in precision closely aligned with the accuracy trends for each model, demonstrating their ability to make correct positive predictions, as shown in Fig. 9. Similar trends are observed in Fig. 10, recall, are observed, indicating the models' effectiveness in capturing positive instances. The Fig. 11, F1-scores reflect the balanced performance of the models, considering both precision and recall. The trends align with accuracy and precision. These comprehensive visualizations aid in understanding the nuances of model performance, enabling informed decisions based on specific evaluation

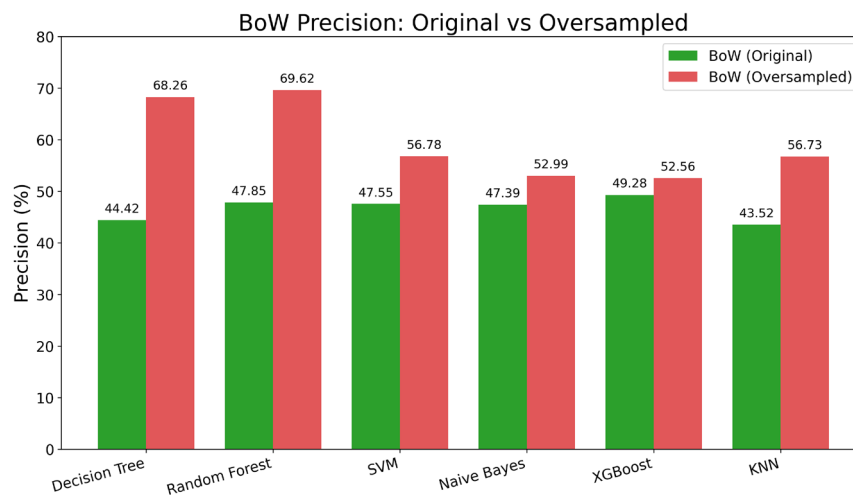


Fig. 9. Precision comparison of different machine learning models for multi-class sentiment classification using BoW in original vs oversampled data.

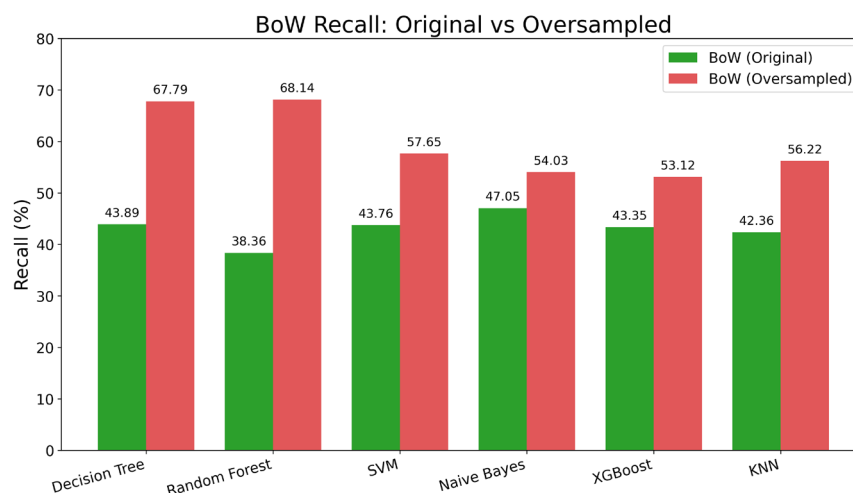


Fig. 10. Recall comparison of different machine learning models for multi-class sentiment classification using BoW in original vs oversampled data.

metrics. TF-IDF continued to provide better results than BOW for most of the other classifiers. SVM achieved the highest accuracy of 65.37% using TF-IDF, leveraging its ability to create more complex separating boundaries. In contrast, using the BoW technique, SVM attained 65.02% accuracy. Oversampling smaller sentiment-rating classes led to significant improvements in model performance. With a balanced class distribution, the Decision Tree delivered the highest accuracy of 68.52% using TF-IDF features.

In contrast, Figs. 12, 13, 14, and 15 depicts set of four bar graphs which offer a comprehensive comparison of machine learning model performances in sentiment classification tasks using Term Frequency-Inverse Document Frequency (TF-IDF). The evaluation metrics included accuracy, precision, recall, and F1-score, both original and oversampled data.

In Fig. 12, decision tree achieves around 59.08% accuracy without oversampling and improves to 68.52% with oversampling. Logistic regression exhibited an accuracy of 65.93% without oversampling and 56.99% with oversampling. Random Forest shows an accuracy of 60.73% without oversampling and 69.06% with oversampling. SVM attained 65.37% accuracy with original and 56.87% with oversampling. Naïve Bayes yielded an accuracy of 63.38% with original and 53.76% with oversampling. XGBoost maintained an accuracy of 64.34% with original and 54.01% with oversampling. KNN achieved an accuracy of 44.67% with original and 52.56% with oversampling, respectively.

The patterns in precision closely aligned with the accuracy trends for each model, as shown in Fig. 13, showcasing their ability to make correct positive predictions. Similar trends in the recall bar graph 14 are observed, indicating the models' effectiveness in capturing positive instances.

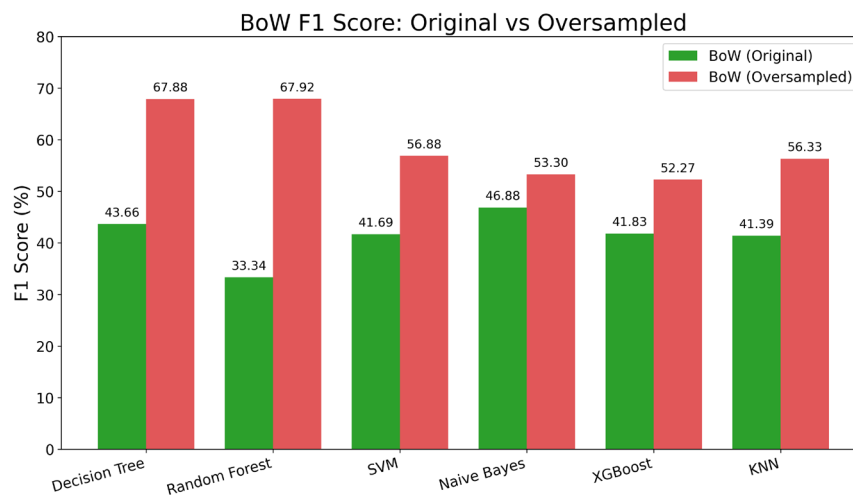


Fig. 11. F1-score comparison of different machine learning models for multi-class sentiment classification using BoW in original vs oversampled data.

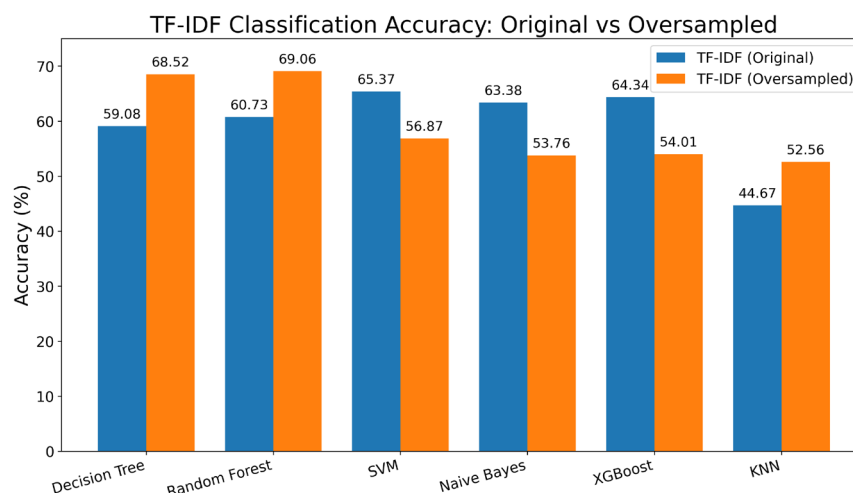


Fig. 12. Accuracy comparison of different machine learning models for multi-class sentiment classification using TF-IDF in original vs oversampled data.

In addition, Fig. 15, F1-scores reflect the balanced performance of the models, considering both precision and recall. The trends aligned with accuracy and precision. These visualizations facilitate a detailed analysis of the models' behavior under the TF-IDF representation, highlighting the impact of oversampling on their classification performance. KNN was again the worst performer for multi-class classification, reaching only 44.67% accuracy with TF-IDF features. This further indicates KNN's unsuitability of KNN for detailed sentiment analysis.

Deep learning models

Figure 16 displays the training and testing accuracy of the proposed model across epochs, revealing a high accuracy level, potentially exceeding 90%. It also showcases the training and testing loss of the proposed model, providing insights into the model's ability to minimize loss during training, with values indicating effective learning.

The confusion matrix shown in Fig. 17 summarizes the model's binary classification performance: True Negatives (Actual Negative, Predicted Negative): 36,889 False Positives (Actual Negative, Predicted Positive): 3,537 False Negatives (Actual Positive, Predicted Negative): 2,790 True Positives (Actual Positive, Predicted Positive): 30,485. This breakdown helps assess the model's accuracy in predicting positive and negative outcomes.

The different ROC curves for the two classes are shown in Fig. 18, which indicates a promising model performance. AUC values of 0.XX and 0.YY demonstrated that both curves performed much better than random guessing, as evidenced by their obvious separation from the dashed line. This implies that the model successfully distinguished between these classes, indicating the need for additional studies in a particular research setting.

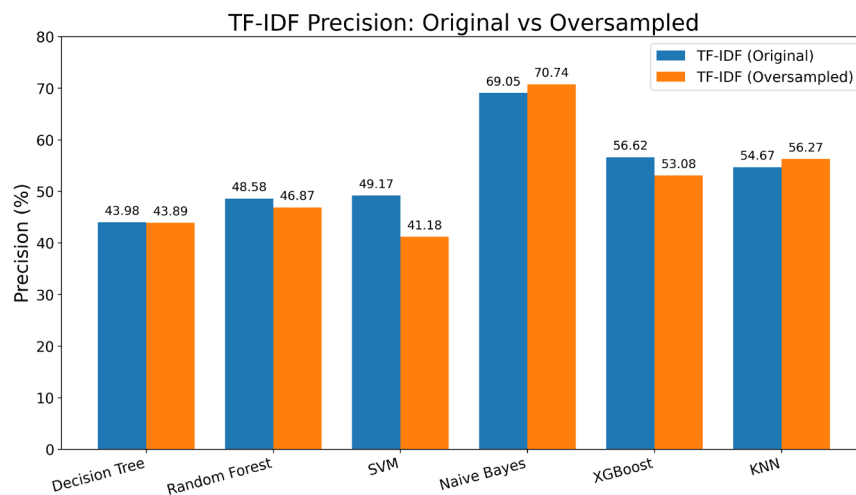


Fig. 13. Precision comparison of different machine learning models for multi-class sentiment classification using TF-IDF in original vs oversampled data.

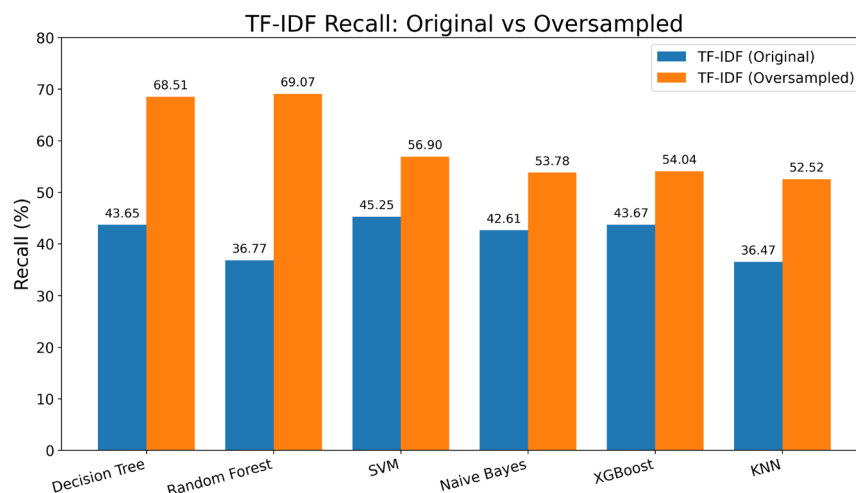


Fig. 14. Recall comparison of different machine learning models for multi-class sentiment classification using TF-IDF in original vs oversampled data.

In Fig. 19, the proposed model's training and testing accuracy are graphically depicted for multi-class classification on balanced data. This Fig. shows the learning dynamics of the model, showing how the accuracy evolved across epochs during the training process. In addition Fig. 19 the Proposed Model's training and testing loss dynamics for multi-class classification on balanced data are visually represented. This Fig. illustrates how the model minimizes loss during the training process, providing insights into its effectiveness in learning from the data. In Fig. 20 the Confusion Matrix for the Proposed Model in multi-class classification on balanced data is presented. This matrix offers a comprehensive breakdown of the model performance, detailing the number of true positives, true negatives, false positives, and false negatives for each class. In Fig. 21 the ROC (Receiver Operating Characteristic) graph for the Proposed Model in multi-class classification on balanced data is displayed. This Fig. provides a visual representation of the model performance across different threshold values, showing the trade-off between the true positive and false positive rates. This offers insights into the ability of the model to discriminate between multiple classes in a balanced dataset.

The DL models also attained strong results, with our proposed model reaching 91.42% test accuracy and 91.35% F1-score using word embeddings. This highlights the capability of DL to learn robust text representations from embeddings for accurate sentiment classification.

The training and testing accuracies of the deep learning models on the original imbalanced dataset for binary and multi-class classifications are listed in Table 5. For binary sentiment prediction, the 1-layer DCNN achieved an exceptional training accuracy of 97.86%, but dropped to 90.60% on the test set, indicating some overfitting. The 2-layer DCNN model attained a slightly lower training accuracy of 96.57%, but matched the test performance of the 1-layer DCNN with 90.71% accuracy. Comparatively, the proposed model demonstrated

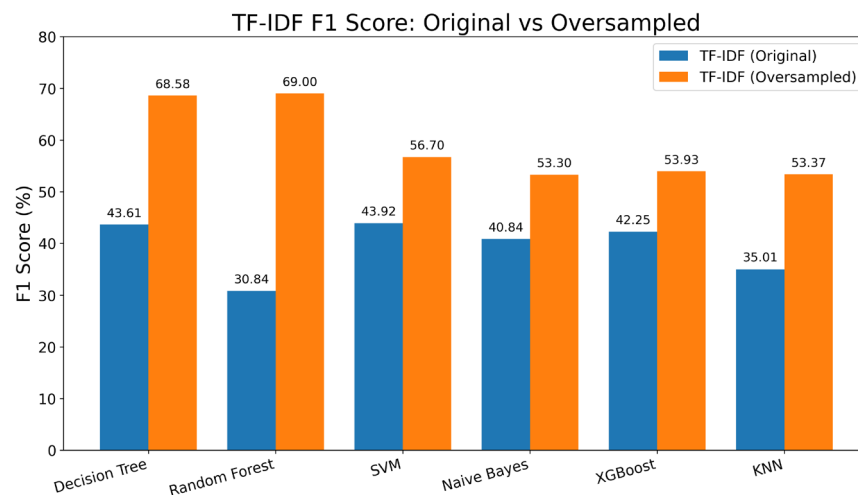


Fig. 15. F1-score comparison of different machine learning models for multi-class sentiment classification using TF-IDF in original vs oversampled data.

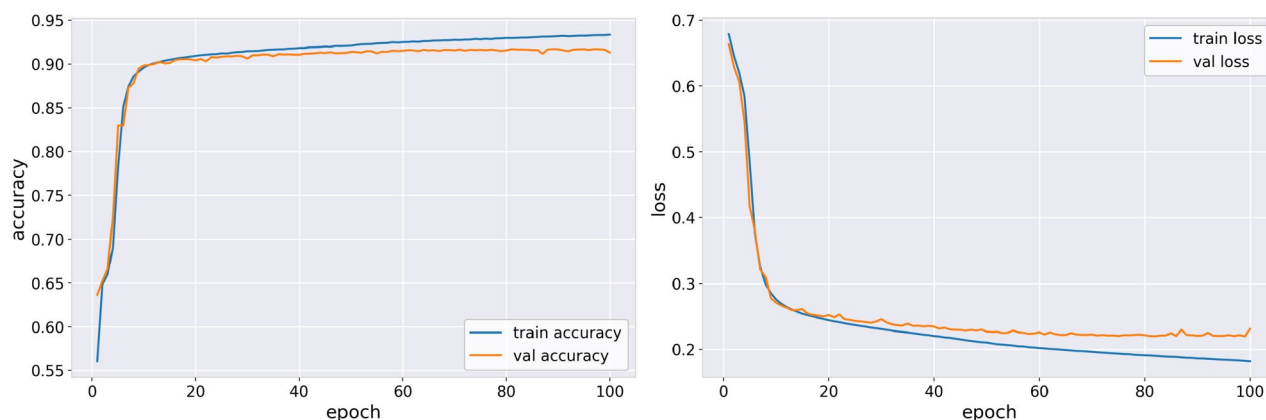


Fig. 16. Proposed model—binary (positive vs. negative): training and val accuracy (left) and loss (right) versus epochs.

balanced outcomes with 93.33% training and 91.42% testing accuracy owing to its robust architecture. For multi-class classification, all models exhibited a larger gap between the training and testing performance owing to class imbalance, with the proposed model achieving the highest test accuracy of 66.76%.

Table 5 shows the training and testing accuracy improvements achieved by the deep learning models when oversampled data were used to mitigate the class imbalance. In the multi-class case, the proposed model attained a high training accuracy of 88.17%, indicating its competence in learning predictive patterns from the balanced classes. More importantly, it sustained a high test accuracy of 78.74% compared to 77.25% for 1-layer DCNN and 76.48% for 2-layer DCNN. The smaller gap between the training and test performances verified the value of oversampling in handling class imbalances.

Table 6 provides a granular view of the performance of the deep learning models on the original imbalanced test data using precision, recall, and F1-score metrics.

For binary sentiment classification, the proposed model achieved the highest precision of 91.29%, indicating that nearly 91% of its positive predictions were correct. Its recall was also strong at 91.43%, indicating that it correctly identified over 91% of the actual positive cases. The F1-score, which balances precision and recall, was 91.35% for the proposed model, again the highest among the deep learning approaches. The 1-layer DCNN attained comparable metrics, whereas the 2-layer DCNN performed marginally worse, likely because of overfitting on the imbalanced data.

In the more challenging multi-class classification scenario, the performance margins were wider. The proposed model achieved 51.87% precision, capturing the highest percentage of correct fine-grained sentiment predictions. Its recall was 49.04%, indicating that approximately half of the actual subtle sentiment instances were classified correctly. The F1-score balanced these metrics at 48.12%, which was significantly higher than those of the 1- and 2-layer DCNNs. These discrepancies highlight the limitations of standard CNNs in learning from imbalanced data.

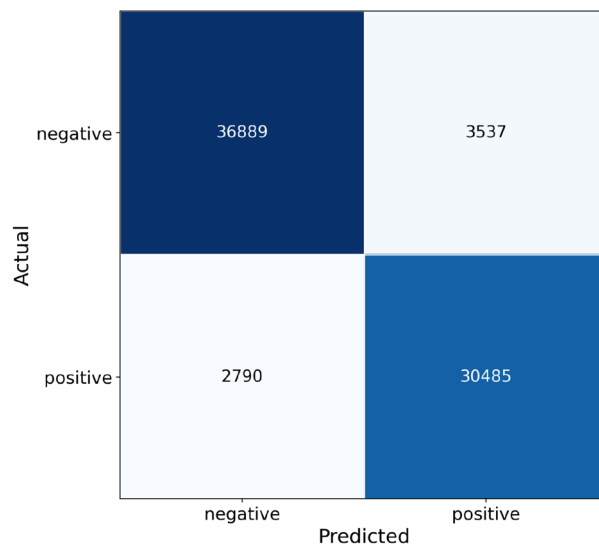


Fig. 17. Proposed Model Confusion Matrix for Binary class (Positive and Negative).

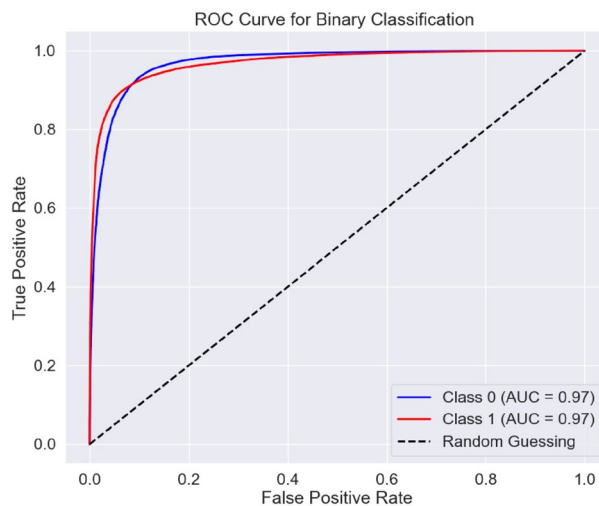


Fig. 18. Proposed Model ROC Graph for Binary class (Positive and Negative).

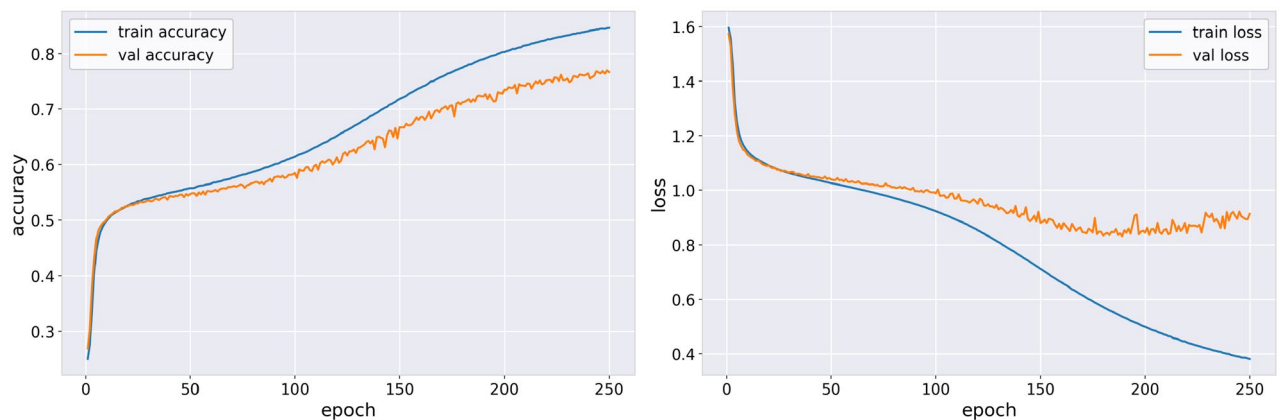


Fig. 19. Proposed model on balanced multi-class data: training and val accuracy (left) and loss (right) versus epochs.

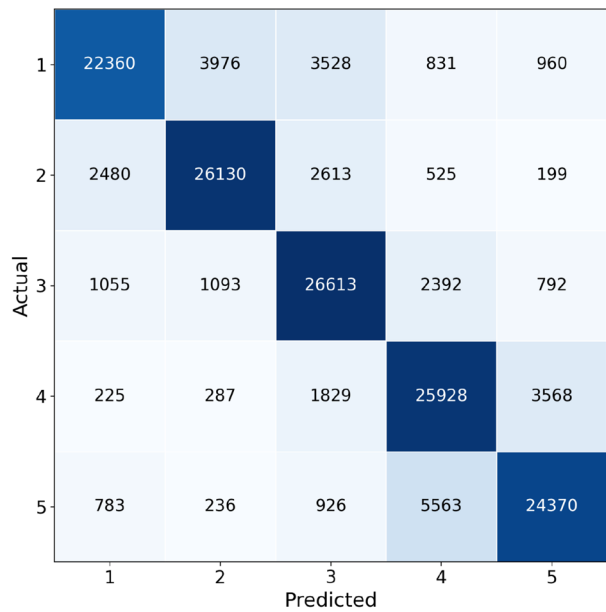


Fig. 20. Proposed Model Confusion Matrix for Multi-class in balanced data.

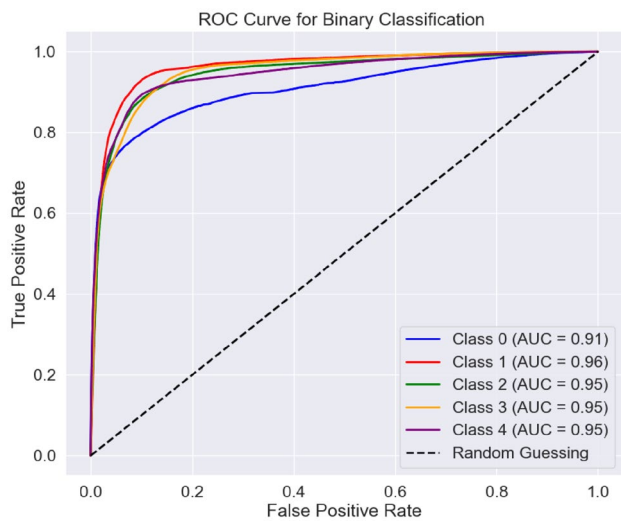


Fig. 21. Proposed Model ROC Graph for Multi-class in balanced data.

DL models	Binary				Multi-class			
	Original		Oversampled		Original		Oversampled	
	Train Acc	Test Acc	Train Acc	Test Acc	Train Acc	Test Acc	Train Acc	Test Acc
1-layer DCNN	97.86	90.60	–	–	78.99	64.14	86.53	77.25
2-layer DCNN	96.57	90.71	–	–	77.35	63.26	86.33	76.48
Proposed model	93.33	91.42	–	–	70.46	66.76	88.17	78.74

Table 5. Train and test accuracy (Acc, %) for original and oversampled datasets across DL models. Dashes (–) indicate results not evaluated/reported.

Finally, Table 6 presents the precision, recall, and F1-scores obtained from oversampling in the multi-class prediction scenario. The proposed model achieved 79.11% precision, capturing the highest percentage of correct fine-grained sentiment predictions from the balanced test data. The recall increased to 78.72%, indicating that approximately 79% of the actual subtle sentiment cases were correctly classified. The F1-score improved to

DL models	Binary (Original)			Multi-class (Original)			Multi-class (Oversampled)		
	P	R	F1	P	R	F1	P	R	F1
1-layer DCNN	90.52	90.52	90.52	48.25	47.29	46.77	77.74	77.24	77.17
2-layer DCNN	90.90	90.37	90.57	48.27	47.92	47.35	77.68	76.46	76.38
Proposed model	91.29	91.43	91.35	51.87	49.04	48.12	79.11	78.72	78.72

Table 6. Test precision (P), recall (R), and F1-score (F1) for original and oversampled datasets across DL models. Binary results are shown only for the original dataset.

Review Text	Task	True Label/Rating	Predicted Label/Rating	Notes
“Did not follow instructions, I asked for breast pcs but received one breast and one leg piece. No one at home likes leg pcs, and overall the taste was poor.”	Binary	Negative	Negative	Correctly classified, strong negative sentiment with multiple complaints
“tasty and fresh piping hot enjoyable”	Binary	Positive	Positive	Correctly classified, clear positive sentiment with strong adjectives
“there is nowhere else such fresh and hot delicious food i have been taking it for so long and always have fun and healthy food the quantity is also the best you can order with your eyes closed”	Binary	Positive	Negative	Misclassified, long review with multiple strong positive cues possibly misinterpreted as exaggeration or noise
“cockroach in the food i do not have food from them regularly but today seeing the cockroach in the food is making me vomit this is ridiculous”	Binary	Negative	Positive	Misclassified, strong negative hygiene-related complaint misinterpreted due to missing explicit negative keywords in training data
“too much small amount”	Multi-class	Very Bad	Very Good	Misclassified, short negative review misinterpreted as very positive due to lack of explicit sentiment cues
“quantity of dal should be little more otherwise everything was good specially beef”	Multi-class	Neutral	Bad	Misclassified, mixed review with minor complaint overshadowing the overall positive context
“very good food i chose this home kitchen randomly but really got a food and packaging please always maintain the same standard allah bless you with all the success”	Multi-class	Very Good	Very Bad	Misclassified, strong positive review completely inverted as very bad, likely due to dataset noise or polarity confusion
“very good standard of foods the pudding is very yummy keep the standard up always”	Multi-class	Good	Very Good	Misclassified, strong positive sentiment slightly exaggerated and pushed into the highest class

Table 7. Representative case analysis of reviews by the proposed H-Food model across binary and multi-class (5-class) sentiment tasks, showing both correctly classified and misclassified examples.

78.72%, balancing the precision and recall. The other deep learning models also exhibited metric improvements, although they were smaller in magnitude. This complements the accuracy trends and substantiates the benefits of oversampling for multi-class sentiment classification.

In summary, the detailed statistical results analysis provides data-backed insights into the significant performance improvements achieved by the proposed model using oversampling to address class imbalance, particularly for fine-grained multi-class sentiment classification.

In summary, the results highlight the efficacy of TF-IDF for sentiment feature extraction and SVM and DL models for a robust classification. Oversampling helped overcome class imbalance issues. The proposed DL model achieved compelling accuracy for both binary and multi-class prediction. KNN is ineffective for detailed sentiment analysis.

Error analysis

The case analysis in Table 7 highlights the strengths and weaknesses of the proposed H-Food model across binary and multiclass sentiment tasks. In the binary setting, the model accurately identified clear positive and negative expressions, indicating that it performed well on simple reviews. However, there are some problems in a multiclass setting. For example, short and unclear complaints (like “too much small amount”) are sometimes read as positive, mixed reviews with minor criticisms are read as negative, and strongly positive reviews may even be switched to the most negative class, which suggests that there is confusion about polarity or possible annotation noise. These examples show how difficult it is to perform fine-grained sentiment classification, where context, subtle cues, and the quality of the data are all very important in determining how well a model works.

Comparative analysis

Table 8 presents a comparative analysis of the performance of classical machine learning models, including baseline CNN architectures and the proposed H-Food model, for both binary and multi-class sentiment classification. Among the machine learning methods, the SVM and TF-IDF features achieved the highest accuracy and F1-score in binary classification, with values being (90.74% and 90.63%, respectively). However, a degradation was seen in the multi-class setting, with results being impacted by class imbalance significantly. Baseline deep learning models with (1-layer and 2-layer DCNNs) performed competitively in binary classification (approximately 90.6–90.7.6.7% accuracy), but revealed moderate overfitting tendencies. When applied to the

Model	Task	Accuracy (%)	F1-score (%)	Notes
Decision Tree (TF-IDF)	Binary	87.34	87.22	Moderate baseline
SVM (TF-IDF)	Binary	90.74	90.63	Best ML model
1-layer DCNN	Binary	90.60	90.52	Some overfitting
2-layer DCNN	Binary	90.71	90.57	Some overfitting
Proposed H-Food	Binary	91.42	91.35	Best overall
Decision Tree (TF-IDF)	Multi-class	68.52 (OS)	67.80	Improved with oversampling
SVM (TF-IDF)	Multi-class	65.37	64.90	Limited by class imbalance
1-layer DCNN	Multi-class	77.25 (OS)	77.17	Better with OS
2-layer DCNN	Multi-class	76.48 (OS)	76.38	Better with OS
Proposed H-Food	Multi-class	78.74 (OS)	78.72	Best overall

Table 8. Comparative performance of machine learning, baseline deep learning, and proposed H-Food model on sentiment classification tasks.

Task	Accuracy	Precision	Recall	F1-score
Binary (2-class)	0.9177 ± 0.0005	0.9178 ± 0.0004	0.9177 ± 0.0005	0.9176 ± 0.0006
Multi-class (5-class, oversampled)	0.7606 ± 0.0064	0.7652 ± 0.0077	0.7606 ± 0.0064	0.7604 ± 0.0060

Table 9. Five-fold cross-validation summary of the proposed H-Food model on binary and multi-class sentiment classification tasks. Values are reported as mean ± standard deviation.

multi-class task with oversampling, CNNs achieved improvements of (76–77% accuracy and F1-score), yet still fell short of the proposed model. The proposed H-Food model consistently exceeded both baselines and other classical approaches, where scores of 91.42% accuracy and 91.35% F1-score in binary classification were achieved, followed by 78.74% accuracy and 78.72% F1-score in multi-class classification with oversampling. These results confirm the robustness of the hybrid architecture in capturing contextual and local patterns, and while also highlighting the importance of oversampling in such cases for mitigating class imbalance in fine-grained sentiment analysis.

The five-fold cross-validation results Table 9 indicates that the proposed H-Food model performs well in binary sentiment classification, attaining accuracy, precision, recall, and F1-scores of approximately 91.8% with small variance, underscoring its robustness and stability across folds. In the more difficult multi-class (five-class, oversampled) setting, the model still does well, with scores of about 76% across all metrics. However, the scores are a little more variable, which shows how hard it is to tell the difference between fine-grained sentiment categories. In general, Table 9 shows that the model is very reliable for binary classification and also does a good job on multi-class tasks.

Limitations

Although this study provides a comprehensive framework for sentiment analysis of restaurant food reviews, still several limitations could be observed. First, although the proposed H-Food model was rigorously validated using five-fold cross-validation, baseline models such as (classical ML and CNNs) were executed only once due to computational resource constraints. Consequently, paired statistical significance tests could not be conducted across the models. Nonetheless, identical preprocessing, feature extraction, and data splits were applied for fairness, and the stability of the proposed model results across folds (with very low variance) provides strong evidence of robustness. Second, a key limitation lies in the translation of Bangla and Banglish reviews into English, where automatic translation may have introduced context loss, semantic drift, or cultural bias, particularly for idiomatic expressions and code-mixed content that are unique to the Bangladeshi context. Although manual spot checks were conducted to enhance translation accuracy, some subtle linguistic nuances may have been overlooked or remain undetected. Future research will extend this study by incorporating native language modeling using transformer-based architectures, such as Bangla-BERT, multilingual BERT, and XLM-R, which can be considered better suited to capture local semantics without relying on translation.

Together, these limitations highlight the need for deeper statistical evaluation and multilingual model development in subsequent studies.

Conclusion and future works

This study proposed a framework for analyzing the sentiment of restaurant reviews taken from food delivery apps in Bangladesh. We solved the problem of insufficient publicly available sentiment resources in a low-resource regional context by creating BDFoodSent, a large-scale annotated dataset containing more than 334,000 reviews. We applied several traditional machine learning methods that used the BoW and TF-IDF feature extraction methods. We found that TF-IDF consistently improved the performance. However, traditional models struggle with multi-class sentiment classification because of class imbalance and their limited ability to capture the contextual nuances. To overcome these problems, we built H-Food, a hybrid deep learning architecture

that combines embeddings, BiLSTM, attention, and convolutional layers, to classify food images. The model performed well on both binary and multi-class sentiment classification tasks, particularly when trained on balanced data using random oversampling. These results show that hierarchical deep learning with appropriate data preprocessing and balancing methods is useful for real-world sentiment analysis. This research not only adds to the body of knowledge on performance, but also provides a reproducible benchmark for regional NLP tasks and shows the importance of creating solutions that are specific to the domain and language used in the dataset.

Future directions include expanding the dataset to incorporate multilingual reviews and multimodal content such as images. We also want to look into transformer-based architectures, such as BERT, Bangla-BERT, and XLM-R, to improve contextual representation and use explainable AI (XAI) methods, such as LIME and SHAP, to make things easier to understand. The goal of these improvements is to help restaurant businesses and food delivery platforms make better and more open decisions by providing them with more information on customer behavior patterns.

Data availability

Data Availability In this study, we have hosted the dataset on Zenodo named as “BDFoodSent: A Large-Scale Sentiment-Labeled Restaurant Review Dataset from Bangladesh” <https://zenodo.org/records/14259962>.

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Author contributions

Md Humaion Kabir Mehedi: Conceptualization, Methodology, Formal Analysis, Writing – Original Draft, Project Administration. Fahmid Al Farid: Software Development, Data Processing, Model Training, Visualization. Ehsanur Rahman Rhythm: Dataset Curation, Validation, Literature Review, Writing - Review & Editing. Farhin Rahman: Data Cleaning, Annotation Coordination, Result Interpretation, Writing - Review & Editing. Khan Md Hasib: Machine Learning Implementation, Hyperparameter Optimization, Experimental Testing. Jia Uddin: Supervision, Manuscript Structuring, Critical Review, High-Level Design Review, Final Approval of the Manuscript. Sarina Mansor: Funding Acquisition, Manuscript Structuring, Critical Review, Ethical Review Oversight, Manuscript Proofreading.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to J.U. or S.M.

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